

ECE321

Electronic circuits

Lecture (07)- Op. Amp_Practical

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- ☐ **Op Amp Characteristics**
- ☐ **Inverting Amplifier**
- ☐ **Non-Inverting Amplifier**

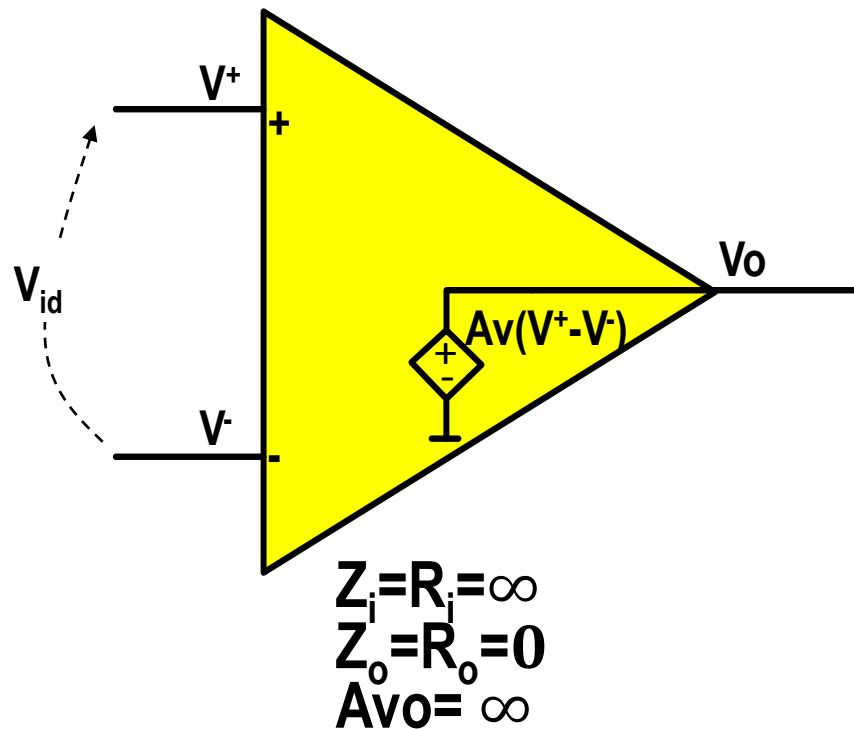


- ☐ **Op Amp Characteristics**
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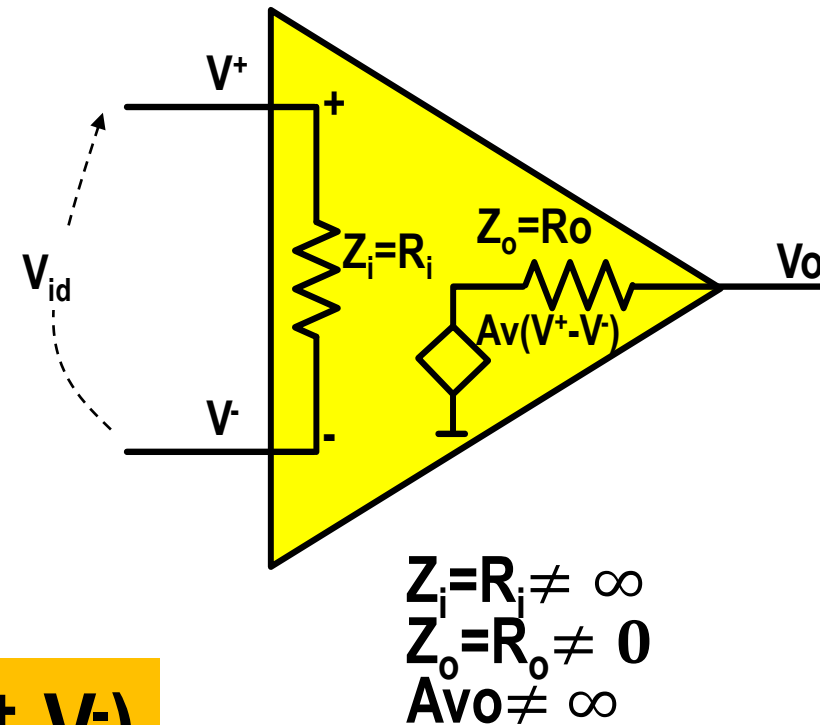
Op Amp Characteristics



Ideal Ch/Cs	Practical Ch/Cs
$I^+ = 0$ $I^- = 0$ $V^+ = V^-$ $A_{v0} = \infty$ // open loop gain is infinite	$I^+ \neq 0$ $I^- \neq 0$ $V^+ \neq V^-$ $A_{v0} \neq \infty$ // open loop gain is finite



$$V_o = A_{v0}(V^+ - V^-)$$



Outlines



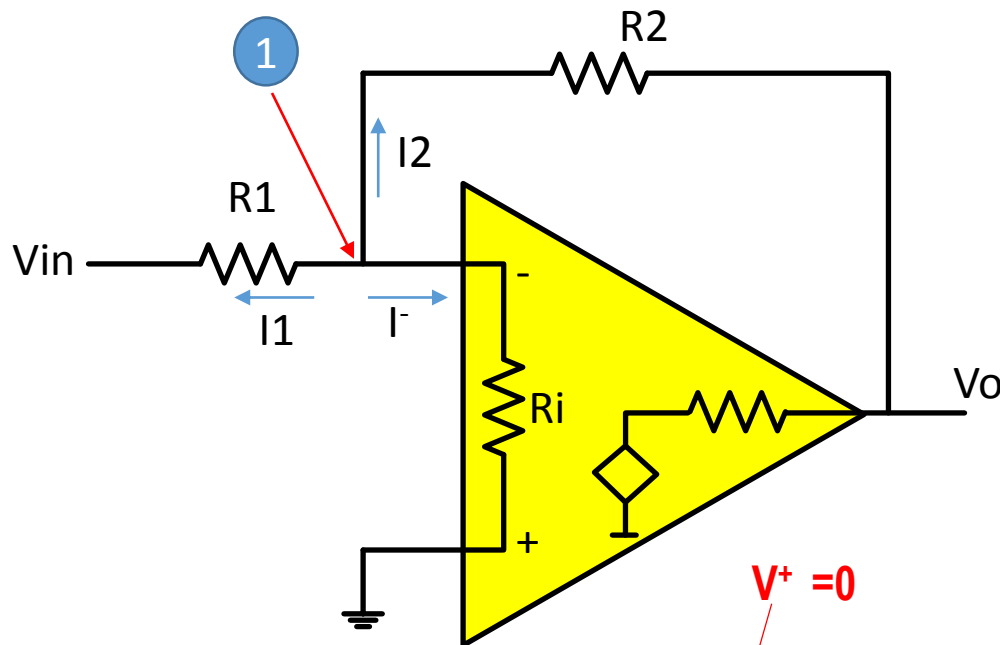
- ☐ Op Amp Characteristics
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Inverting Amplifier (1/4)



- Derive an expression for close loop gain (A_v) for inverting amplifier which has finite open loop gain ($A_{v_o} \neq \infty$)

Ans.



For Practical amplifier

$$I^+ \neq I^- \neq 0$$

$$V^+ \neq V^-$$

$$V_o = A_{v_o}(V^+ - V^-)$$

$$V_o = -A_{v_o} V^- \rightarrow \textcircled{1}$$

Take node @ point (1)

$$I^- + I_1 + I_2 = 0$$

$$\frac{V^-}{R_i} + \frac{V^- - V_{in}}{R_1} + \frac{V^- - V_o}{R_2} = 0$$

$$V^- \left(\frac{1}{R_i} + \frac{1}{R_1} + \frac{1}{R_2} \right) - \frac{V_o}{R_2} - \frac{V_{in}}{R_1} = 0$$

Where typically $R_i \gg R_1$ and R_2

$$V^- \left(\frac{R_1 + R_2}{R_1 * R_2} \right) - \frac{V_o}{R_2} - \frac{V_{in}}{R_1} = 0 \rightarrow \textcircled{2}$$

From (1) and (2)

$$\frac{-V_o}{A_{v_o}} * \left(\frac{R_1 + R_2}{R_1 * R_2} \right) - \frac{V_o}{R_2} - \frac{V_{in}}{R_1} = 0$$

$$\frac{V_o}{V_{in}} = \frac{-R_2}{R_1} * \left(\frac{1}{1 + \frac{R_1 + R_2}{R_1 * A_{v_o}}} \right)$$



Inverting Amplifier (2/4)



Notes

$$\frac{V_o}{V_{in}} = \frac{-R_2}{R_1} * \left(\frac{1}{1 + \frac{R_1 + R_2}{R_1 * A_{V_o}}} \right)$$

□ If $A_{V_o} = \infty$, that led $A_v = \frac{V_o}{V_{in}} = \frac{-R_1}{R_2}$

□ If $A_{V_o} \neq \infty$ and a little bite small, that led to an error in calculating close loop gain

□ Ex., if $R_2 = 100\text{k}\Omega$, $R_1 = 10\text{k}\Omega$, $A_{V_o} = 100000$ (100dB), find error occurred in close loop gain

Ideal Op. amp	Practical Op. amp
$A_v = \frac{V_o}{V_{in}} = \frac{-R_1}{R_2} = - (100\text{k}/10\text{k}) = -10$	$A_v = \frac{V_o}{V_{in}} = \frac{-R_2}{R_1} * \left(\frac{1}{1 + \frac{R_1 + R_2}{R_1 * A_{V_o}}} \right)$ $A_v = -9.989$

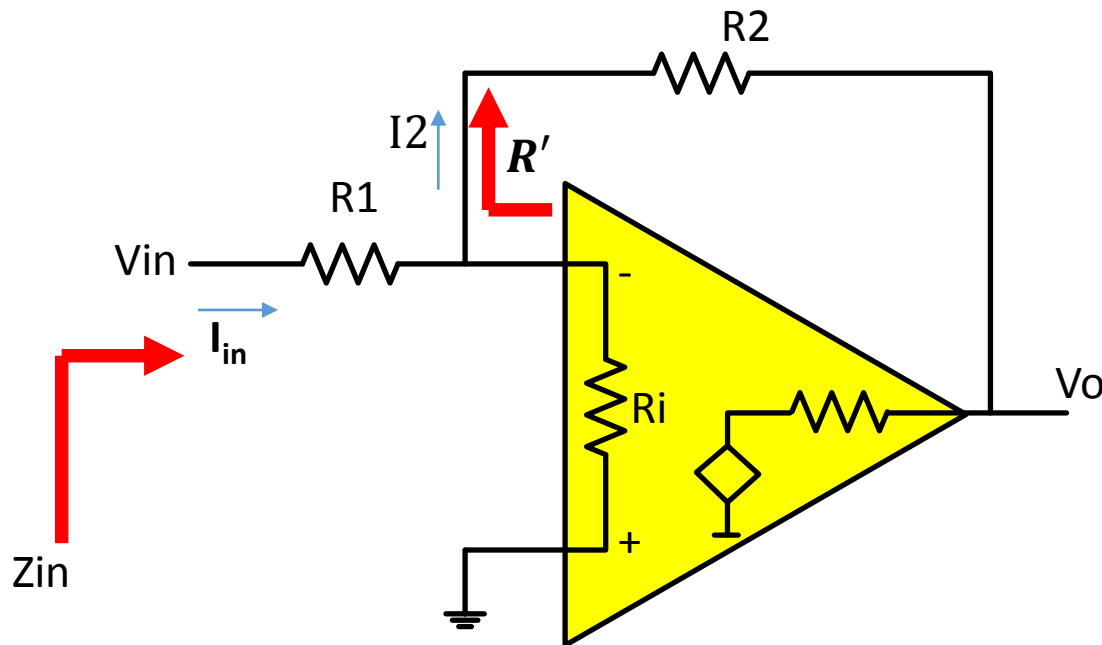
$$\text{Relative Error (\%)} = \left| \frac{A_{v(\text{ideal})} - A_{v(\text{Practical})}}{A_{v(\text{ideal})}} \right| = \left| \frac{-10 - (-9.989)}{-10} \right| = \sim 0.1\%$$

Inverting Amplifier (3/4)



- Derive an expression for Z_{in} for inverting amplifier which has finite open loop gain ($A_{Vo} \neq \infty$)

Ans.



$$Z_{in} = \frac{V_{in}}{I_{in}}$$

$$Z_{in} = R1 + Ri || R'$$

To get R'

$$R' = \frac{V^-}{I_2} = \frac{V^-}{\frac{V^- - V_o}{R2}} = \frac{V^-}{V^- - V_o} R2$$

$$V_o = A_{vo}(V^+ - V^-)$$

$$V_o = -A_{vo} V^-$$

$$R' = \frac{V^-}{V^- + A_{vo} * V^-} R2$$

$$R' = \frac{R2}{1 + A_{vo}}$$

$$Z_{in} = R1 + Ri || \frac{R2}{1 + A_{vo}}$$

Note

If $A_{Vo} = \infty$, that led $Z_{in} = \frac{V_o}{V_{in}} = R1$

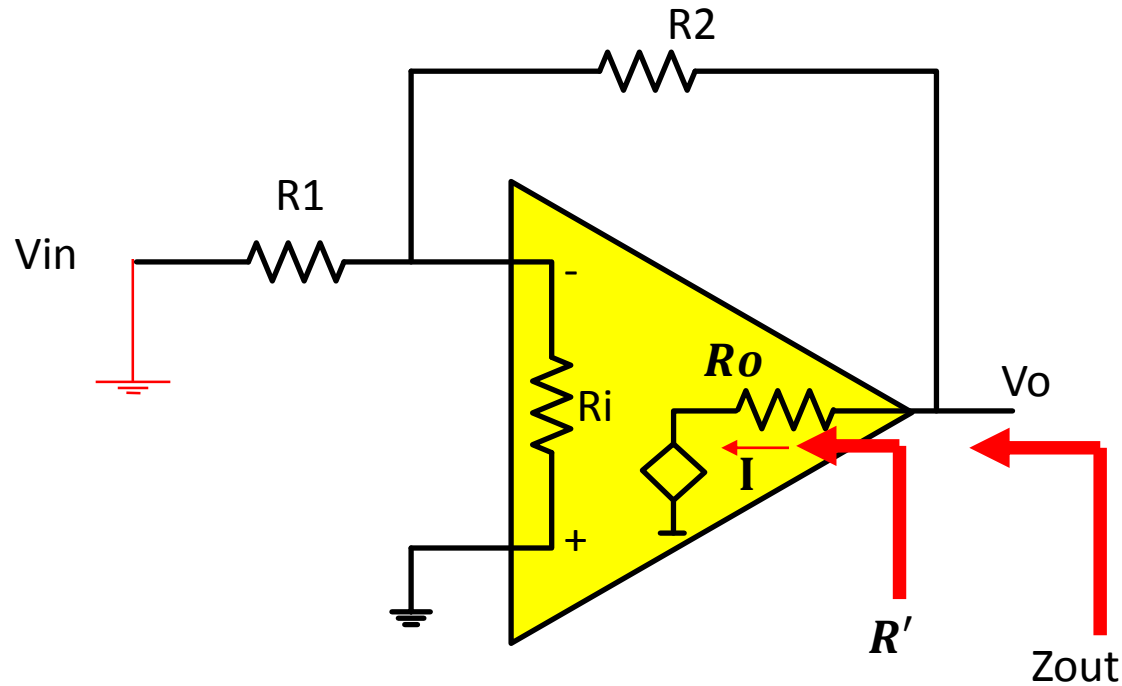
If $A_{Vo} \neq \infty$ and a little bite small, that led to an error in calculating Z_{in}

Inverting Amplifier (4/4)



- Derive an expression for Z_{out} for inverting amplifier which has finite open loop gain ($A_{V_o} \neq \infty$)

Ans.



$$Z_{out} = \frac{V_o}{I_o} @ V_{in}=0$$

$V_o =$

$$Z_{out} = [R2 + (R1 || Ri)] || R'$$

To get R'

$$R' = \frac{V_o}{I} = \frac{V_o}{\frac{V_o - A_{V_o}(V^+ - V^-)}{R_o}} = \frac{V_o}{\frac{V_o + A_{V_o}(V^-)}{R_o}}$$

$$V^- = V_o \frac{R1}{R1 + R2}$$

$$R' = \frac{V_o}{\frac{V_o + A_{V_o}(V_o \frac{R1}{R1 + R2})}{R_o}} = \frac{R_o}{1 + A_{V_o}(\frac{R1}{R1 + R2})}$$

$$Z_{out} = [R2 + (R1 || Ri)] || \frac{R_o}{1 + A_{V_o}(\frac{R1}{R1 + R2})}$$

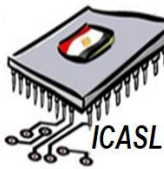
Note

- If $A_{V_o} = \infty$, $R_i = \infty$ that led $Z_{out} = 0$
- If $A_{V_o} \neq \infty$ and a little bite small, that led to an error in calculating Z_{out}



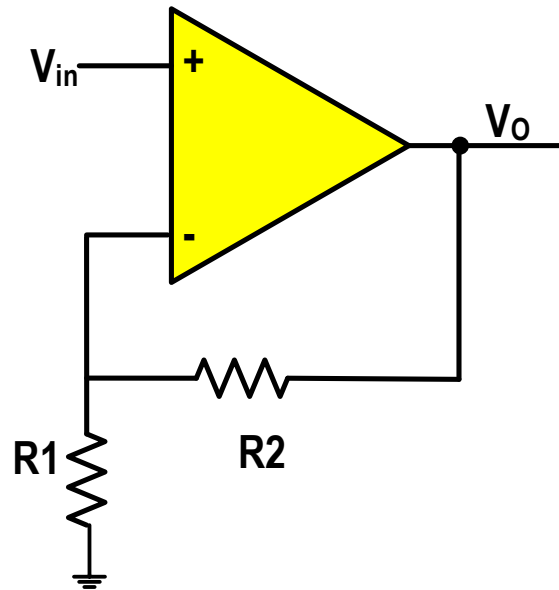
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Non-Inverting Amplifier



- Derive an expression for close loop gain (A_v), Z_{in} , and Z_{out} for non-inverting amplifier which has finite open loop gain ($A_{v_o} \neq \infty$)

Ans.



Ideal Op. Amp

$$A_v = \left(1 + \frac{R_2}{R_1}\right)$$

$$Z_{in} = \infty$$

$$Z_{out} = \infty$$

Practical Op. Amp

$$A_v = \left(1 + \frac{R_2}{R_1}\right) \left(\frac{1}{1 + \frac{R_1 + R_2}{R_1 * A_{v_o}}}\right)$$

$$Z_{in} = R_1 + R_i \parallel \frac{R_2}{(1 + A_{v_o}) - A_{v_o} \left(1 + \frac{R_1 + R_2}{R_1 * A_{v_o}}\right)}$$

$$Z_{out} = \frac{R_o}{1 + A_{v_o} \frac{R_1}{R_1 + R_2}} \parallel [R_2 + (R_1 \parallel R_i)]$$

The detailed answer is depicted in the lecture notes

Thanks!